Charge and the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain charge interactions and trace an electric car's energy transfers.

SUCCESS CRITERIA - I can...

1. Predict attraction or repulsion from charge signs.
2. Distinguish charge from energy.
3. Trace energy from a battery to moving wheels.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

An ideal blackbody's kelvin temperature becomes 3 times as large. How does its peak wavelength change?

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $E_{electrical} = E_{mechanical} + E_{thermal}$ + other transfers. Energy accounting for a motor with no net energy storage. A battery transfers energy; charge already exists in the wires.

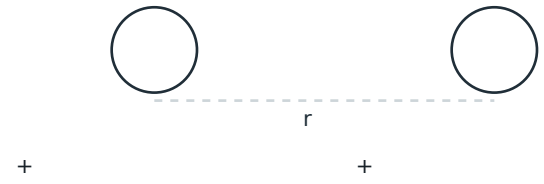
Worked reasoning: Rubbing can transfer electrons between materials. If one isolated object gains charge -3 units and another loses the same electrons, their charge changes add to zero.

Your turn: Draw before-and-after charge accounts for both objects. Distinguish transfer from creation of charge.

Marzano check - circle: ① ② ③

03 Look and explain

Draw force arrows for two positive charges, then for a positive and a negative charge.



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Build the solution

Put these energy forms in order: car motion, chemical energy in battery, electrical transfer, motor rotation. Name one less useful transfer.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Use the idea

A student says the motor uses up the current so less returns to the battery. Explain the mistake for a steady series circuit.

Marzano check - circle: 1 2 3

06 Choose a method

Two wave displacements at one point are -4 mm and +8 mm. Find the resultant and state whether they reinforce or partly cancel.

Marzano check - circle: 1 2 3

07 Exit check

Predict the interaction between two negative charges.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Draw one attraction between opposite charges and one repulsion between like charges. Explain whether attraction alone proves both objects carry opposite net charges.

Marzano check - circle: 1 2 3